

MATH 5152: Numerical Methods for Data Science

Homework 6

Due: Monday, March 9 at 11:59 am

Submission Instructions: Submit a single PDF file named `LASTNAME-hw6.pdf`. All solutions must be typed in L^AT_EX and uploaded to Canvas before the deadline. No credit will be given for answers without supporting work — you are expected to show all steps clearly. The use of any AI tools (including OpenAI models) to generate solutions is strictly prohibited.

1. Compute the gradient ∇f and Hessian $\nabla^2 f$ for the following functions.

(a) $f_1(x, y) = e^{xy^2 + \sin(x)}$

$$\nabla f_1 = \begin{pmatrix} e^{xy^2 + \sin x} (y^2 + \cos x) \\ 2xy e^{xy^2 + \sin x} \end{pmatrix}$$

$$\begin{aligned} f_{xx} &= \frac{\partial}{\partial x} \left[e^{xy^2 + \sin x} (y^2 + \cos x) \right] = e^{xy^2 + \sin x} (y^2 + \cos x)^2 - \sin(x) e^{xy^2 + \sin x} = \\ &= e^{xy^2 + \sin x} [(y^2 + \cos x)^2 - \sin x] \end{aligned}$$

$$\begin{aligned} f_{yy} &= \frac{\partial}{\partial y} \left[2xy e^{xy^2 + \sin x} \right] = 2x e^{xy^2 + \sin x} + 4x^2 y^2 e^{xy^2 + \sin x} = \\ &= e^{xy^2 + \sin x} (2x + 4x^2 y^2) \end{aligned}$$

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y} \left[e^{xy^2 + \sin x} (y^2 + \cos x) \right] = 2y e^{xy^2 + \sin x} + 2xy(y^2 + \cos x) e^{xy^2 + \sin x} = \\ &= e^{xy^2 + \sin x} [2y + 2xy(y^2 + \cos x)] \end{aligned}$$

$$\nabla^2 f_1 = e^{xy^2 + \sin x} \begin{pmatrix} (y^2 + \cos x)^2 - \sin x & 2y + 2xy(y^2 + \cos x) \\ 2y + 2xy(y^2 + \cos x) & 2x + 4x^2 y^2 \end{pmatrix}$$

(b) $f_2(x, y) = \ln(e^{x+y} + e^{x-y})$

$$\frac{\partial f_2}{\partial x} = \frac{e^{x+y} + e^{x-y}}{e^{x+y} + e^{x-y}} = 1$$

$$\frac{\partial f_2}{\partial y} = \frac{e^{x+y} - e^{x-y}}{e^{x+y} + e^{x-y}}$$

$$\nabla f_2 = \begin{pmatrix} 1 \\ \frac{e^{x+y} - e^{x-y}}{e^{x+y} + e^{x-y}} \end{pmatrix}$$

$$f_{yy} = \frac{(e^{x+y} + e^{x-y})^2 - (e^{x+y} - e^{x-y})^2}{(e^{x+y} + e^{x-y})^2} = 1 - \frac{(e^{x+y} - e^{x-y})^2}{(e^{x+y} + e^{x-y})^2}$$

$$\nabla^2 f_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 - \frac{(e^{x+y} - e^{x-y})^2}{(e^{x+y} + e^{x-y})^2} \end{pmatrix}$$

$$(c) \quad f_3(w) = \sum_{i=1}^n \log(1 + e^{-y_i w^\top x_i}) + \frac{\lambda}{2} \|w\|_2^2$$

$$\frac{\partial}{\partial w} \log(1 + e^{-y_i w^\top x_i}) = \frac{-y_i x_i e^{-y_i w^\top x_i}}{1 + e^{-y_i w^\top x_i}}$$

Then we have: $\frac{\partial}{\partial w} \frac{\lambda}{2} \|w\|_2^2 = \lambda w$. Therefore:

$$\nabla f_3 = \sum_{i=1}^n \frac{-y_i x_i e^{-y_i w^\top x_i}}{1 + e^{-y_i w^\top x_i}} + \lambda w$$

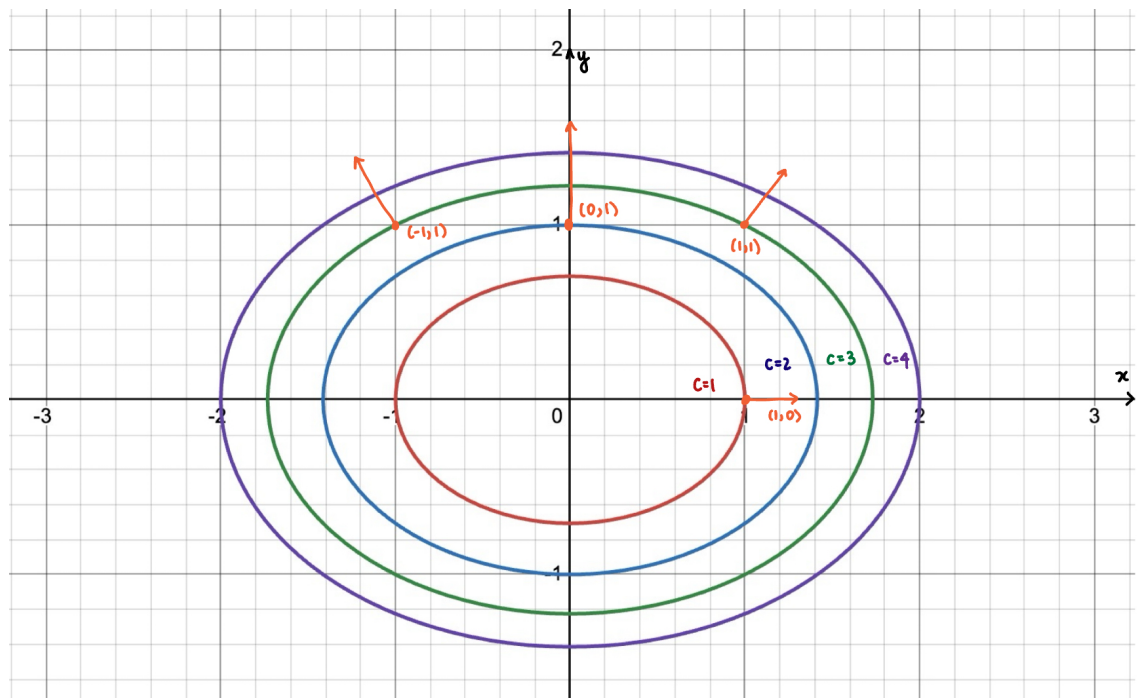
$$\begin{aligned} \frac{\partial}{\partial w^\top} \left[\frac{-y_i x_i e^{-y_i w^\top x_i}}{1 + e^{-y_i w^\top x_i}} \right] &= \frac{y_i^2 x_i x_i^\top e^{-y_i w^\top x_i} (1 + e^{-y_i w^\top x_i}) - y_i^2 x_i x_i^\top e^{-2y_i w^\top x_i}}{(1 + e^{-y_i w^\top x_i})^2} = \\ &= \frac{y_i^2 x_i x_i^\top e^{-y_i w^\top x_i}}{(1 + e^{-y_i w^\top x_i})^2} \end{aligned}$$

This is the loss function for logistic regression! So...we use that auxiliary information and the fact that $y_i \in \{-1, +1\}$ (binary logistic regression with two classes/outcomes, so we have $y_i^2 = 1$) to perform the following substitution:

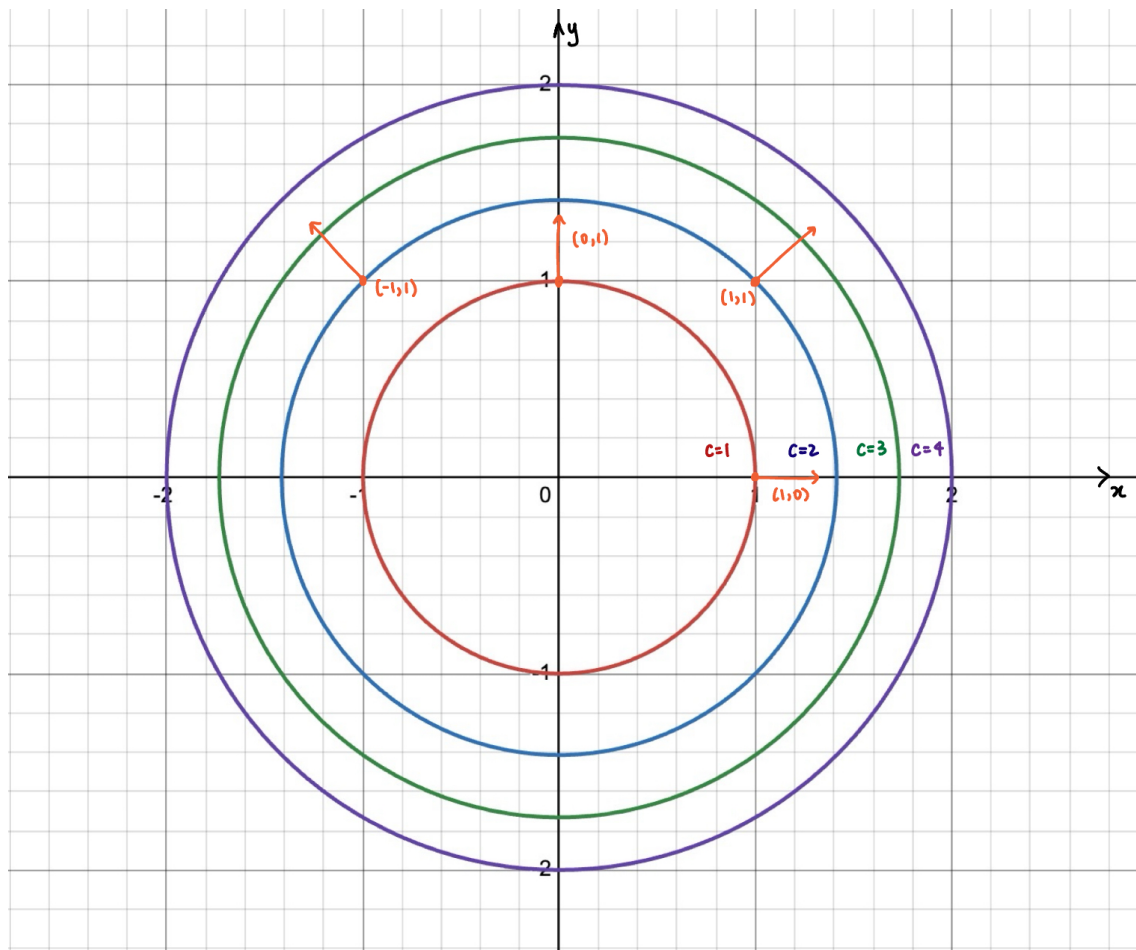
$$\nabla^2 f_3 = \sum_{i=1}^n \frac{e^{-y_i w^\top x_i}}{(1 + e^{-y_i w^\top x_i})^2} x_i x_i^\top + \lambda I$$

2. Consider $f(x, y) = x^2 + 2y^2$.

(a) Sketch the level curves of f for $c = 1, 2, 3, 4$.



- (b) Compare with $f(x, y) = x^2 + y^2$. How do the level curves shapes differ?
- (c) Sketch the level curves of f for $c = 1, 2, 3, 4$.



For $x^2 + y^2 = c$, the level curves are circles of radius $\sqrt{c}$, meaning they are symmetric in x and y . For $x^2 + 2y^2 = c$, the larger coefficient on y^2 squeezes the curves in the y -direction by a factor of $\sqrt{2}$, producing ellipses.

- (d) Compute and draw the gradient vectors on your sketch at the points

$$(1, 0), \quad (0, 1), \quad (1, 1), \quad (-1, 1).$$

Comment on the direction and relative magnitude of the gradient vectors.
The gradient of f is:

$$\nabla f = \begin{pmatrix} 2x \\ 4y \end{pmatrix}$$

Note that $\nabla f(1, 0) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, $\nabla f(0, 1) = \begin{pmatrix} 0 \\ 4 \end{pmatrix}$, $\nabla f(1, 1) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, and $\nabla f(-1, 1) = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$. The gradient vectors are drawn above in the figure, although it should be

noted that the magnitudes (2, 4, $2\sqrt{5}$, and $2\sqrt{5}$, respectively) are not accurately represented above (I scaled them to fit in the image).

3. Let $f(x, y) = e^{x^2+y^2} - 4xy$.

(a) Compute the gradient $\nabla f(x, y)$.

$$\frac{\partial f}{\partial x} = 2xe^{x^2+y^2} - 4y, \quad \frac{\partial f}{\partial y} = 2ye^{x^2+y^2} - 4x$$

$$\nabla f = \begin{pmatrix} 2xe^{x^2+y^2} - 4y \\ 2ye^{x^2+y^2} - 4x \end{pmatrix}$$

(b) Compute the Hessian $\nabla^2 f(x, y)$ explicitly.

$$f_{xx} = \frac{\partial}{\partial x} [2xe^{x^2+y^2} - 4y] = 2e^{x^2+y^2} + 4x^2e^{x^2+y^2} = (2 + 4x^2)e^{x^2+y^2}$$

$$f_{yy} = \frac{\partial}{\partial y} [2ye^{x^2+y^2} - 4x] = 2e^{x^2+y^2} + 4y^2e^{x^2+y^2} = (2 + 4y^2)e^{x^2+y^2}$$

$$f_{xy} = \frac{\partial}{\partial y} [2xe^{x^2+y^2} - 4y] = 4xye^{x^2+y^2} - 4$$

$$\nabla^2 f = \begin{pmatrix} (2 + 4x^2)e^{x^2+y^2} & 4xye^{x^2+y^2} - 4 \\ 4xye^{x^2+y^2} - 4 & (2 + 4y^2)e^{x^2+y^2} \end{pmatrix}$$

(c) Evaluate both ∇f and $\nabla^2 f$ at $(0, 0)$ and at $(1, 1)$.

$$\nabla f(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \nabla^2 f(0, 0) = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix}$$

At $(1, 1)$: We have e^2 , so:

$$\nabla f(1, 1) = \begin{pmatrix} 2e^2 - 4 \\ 2e^2 - 4 \end{pmatrix}, \quad \nabla^2 f(1, 1) = \begin{pmatrix} 6e^2 & 4e^2 - 4 \\ 4e^2 - 4 & 6e^2 \end{pmatrix}$$

(d) For each point, determine:

- whether the Hessian is symmetric,
 - whether it is positive definite, negative definite, or indefinite.
- The Hessian is symmetric at both points, as $f_{xy} = f_{yx}$.

At $(0, 0)$, the characteristic polynomial is:

$$\det(H - \lambda I) = (2 - \lambda)^2 - 16 = \lambda^2 - 4\lambda - 12 = (\lambda - 6)(\lambda + 2) = 0$$

The eigenvalues are $\lambda_1 = 6 > 0$ and $\lambda_2 = -2 < 0$, so the Hessian is **indefinite**.

At (1, 1): The matrix has equal diagonal entries $6e^2$ and off-diagonal entries $4e^2 - 4$, so the eigenvalues are:

$$\lambda = 6e^2 \pm (4e^2 - 4)$$

giving $\lambda_1 = 10e^2 - 4$ and $\lambda_2 = 2e^2 + 4$. Since $e^2 > 0$, both eigenvalues are strictly positive, so the Hessian is **positive definite** at $(1, 1)$.

4. Let $x \in \mathbb{R}^n$, $Q \in \mathbb{R}^{n \times n}$, and $c \in \mathbb{R}^n$.

(a) For $f(x) = x^\top Qx$, show that $\nabla_x(x^\top Qx) = (Q + Q^\top)x$.

Hints: Expand $f(x) = \sum_{i,j} Q_{ij}x_i x_j$ and differentiate component-wise. and use the product rule $\frac{\partial}{\partial x_k}(x_i x_j) = \delta_{ik}x_j + \delta_{jk}x_i$, where δ_{ij} is the Kronecker delta.

Differentiating with respect to x_k , we get:

$$\frac{\partial f}{\partial x_k} = \sum_{i,j} Q_{ij} \frac{\partial(x_i x_j)}{\partial x_k} = \sum_{i,j} Q_{ij} (\delta_{ik}x_j + \delta_{jk}x_i)$$

Splitting the sum using the Kronecker delta to collapse the indices:

$$= \sum_j Q_{kj}x_j + \sum_i Q_{ik}x_i = (Qx)_k + (Q^\top x)_k$$

Since this holds for all $k = 1, \dots, n$, we have:

$$\nabla_x(x^\top Qx) = Qx + Q^\top x = (Q + Q^\top)x \quad \square$$

In the special case where Q is symmetric ($Q = Q^\top$), this reduces to $2Qx$.

(b) For $g(x) = c^\top x$, show that $\nabla_x(c^\top x) = c$.

Differentiating with respect to x_k :

$$\frac{\partial g}{\partial x_k} = \sum_i c_i \frac{\partial x_i}{\partial x_k} = \sum_i c_i \delta_{ik} = c_k$$

Since this holds for all $k = 1, \dots, n$:

$$\nabla_x(c^\top x) = c \quad \square$$